

PRACTICAL NO. 9

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses technique for the page Optimal page replacement technique references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

CODE:

```
MINGW64:/d/OS_CM24112/OSLab_CM24112
GNU nano 8.5 optimal.c Modified
#include <stdio.h>

int main() {
    int frames = 3;
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;

    int memory[3];
    int i, j, k;
    int hit = 0, fault = 0;

    // Initialize
    for(i = 0; i < frames; i++) {
        memory[i] = -1;
    }

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check HIT
        for(j = 0; j < frames; j++) {
            if(memory[j] == pages[i]) {
                hit++;
                found = 1;
                break;
            }
        }

        // If FAULT
        if(!found) {
            int index = -1, farthest = i + 1;

            for(j = 0; j < frames; j++) {
                int k;
                for(k = i + 1; k < n; k++) {
                    if(memory[j] == pages[k]) {
                        if(k > farthest) {
                            farthest = k;
                            index = j;
                        }
                    }
                }
                break;
            }
        }
    }
}
```

```

        // If page not found in future
        if(k == n) {
            index = j;
            break;
        }
    }

    // If all pages will be used soon
    if(index == -1) index = 0;

    memory[index] = pages[i];
    fault++;
}

printf("Total Hits = %d\n", hit);
printf("Total Faults = %d\n", fault);

float hit_per = (float)hit / n * 100;
float fault_per = (float)fault / n * 100;

printf("Hit Percentage = %.2f%%\n", hit_per);
printf("Fault Percentage = %.2f%%\n", fault_per);

return 0;
}

```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

OUTPUT:

 MINGW64:/d/OS_CM24112/OSLab_CM24112

```

Shars@Shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112 (main)
$ nano optimal.c

```

```

Shars@Shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112 (main)
$ gcc optimal.c -o optimal
./optimal
Total Hits = 5
Total Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

```

```

Shars@Shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112 (main)
$

```